

DS LAB FINAL-Explanation of approach

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Chapter 1

Digital Clock System Design Report

Overview

This digital clock module is designed in Verilog and performs multiple timing functions using a 1Hz input clock. It tracks time and date, supports 12-hour and 24-hour display modes, includes a countdown timer, and allows for an alarm to be set. The design uses synchronous logic and internal registers to manage time progression, event triggers, and user inputs.

Timekeeping and Calendar Logic

The core of the module maintains and updates seconds, minutes, and hours each second. When seconds reach 59, they reset and minutes are incremented. Similarly, minutes roll over into hours, and hours into days. The module includes logic to correctly increment the date, considering the varying number of days in each month and handling leap years by checking if the year is divisible by 4. When the end of the year is reached, the month and year increment accordingly.

Time Format Conversion

The module supports both 12-hour and 24-hour time displays. A mode input determines the format. In 24-hour mode, the output hour is shown directly. In 12-hour mode, the output hour is adjusted accordingly with AM or PM indication. Special handling ensures midnight and noon are displayed correctly (12 AM and 12 PM respectively), and times after noon are converted to 1–11 PM.

Manual Time Setting

The module allows the user to manually set the time by asserting a control signal. When active, this signal causes the internal time registers (hours, minutes, seconds) to be overwritten by external input values. This feature is useful for initializing the clock or adjusting it when necessary.

Countdown Timer Logic

A countdown timer is implemented with minute and second registers. The timer starts on a rising edge signal. Once started, it decrements seconds and minutes each clock cycle. When the countdown reaches zero, a buzzer is activated for one second to signal completion. Internal flags ensure the buzzer only pulses once per countdown session.

Alarm Functionality

An alarm system compares the current time to a user-defined alarm time. When the time matches, and if the alarm is enabled, the alarm buzzer is activated for one second. A pulse flag is used to ensure the alarm only triggers once per matching condition to avoid repeated buzzing within the same second.

Conclusion

This module integrates digital timekeeping, alarm and timer functionalities in a compact design. Its handling of calendar progression, dual-format display, and user interactivity makes it suitable for FPGA-based clock applications or embedded systems requiring time-aware operations.

Chapter 2

Traffic Light Design Report

Introduction

This report describes the design and operation of a digital traffic signal controller for a T-junction, as per the problem statement. The controller manages two independent traffic flows (T1 and T2), pedestrian crossings, and emergency ambulance scenarios using a finite state machine (FSM) approach. The design ensures safety, modularity, and real-world applicability.

Design Ideology

- **FSM Approach:** Each traffic light (T1, T2) is modeled as an independent FSM with states: RED, GREEN, YELLOW. This modular approach ensures clarity and scalability.
- **Pedestrian Integration:** Pedestrian walk signals (T1_WALK, T2_WALK) are tied to the RED and YELLOW states of their respective traffic lights and are disabled during emergencies for safety.
- **Emergency Handling:** Emergency signals (Emergency_Left, Emergency_Right) allow ambulances to stop traffic as needed. The system resumes from the exact state and time after the emergency, ensuring continuity.
- **Timing:** Each state has a fixed duration: RED for 60s, GREEN for 30s, YELLOW for 5s. A 1Hz clock is used for all timing.

FSM Structure and Operation

- **State Encoding:** States are encoded as RED = 2'b00, GREEN = 2'b01, YELLOW = 2'b10.
- **Counters:** Each FSM has its own counter (T1_counter, T2_counter) incremented every second.
- **Normal Operation:** Each FSM cycles through RED (60s) → GREEN (30s) → YELLOW (5s) → RED.
- **Pedestrian Walk Logic:** It is active HIGH when traffic light is RED, LOW during emergencies
- **Buzzer:** The buzzer is activated during the last 5 seconds of the pedestrian walk period.
- **Emergency Handling:** When an emergency occurs:
 - Emergency_Left: T1 is forced to RED for 10s, T2 continues as normal.
 - Emergency_Right: Both T1 and T2 are forced to RED for 10s.
 - During emergency, the normal counters are paused. After emergency, FSM resumes from the previous state and counter value.

Code Hierarchy and Monitoring

1. **State Encoding:** Defined using parameters for clarity.
2. **Register Declarations:** Separate registers for current/next state and counters for T1 and T2.
3. **State Update:** Clock-triggered always block updates the present state and counters.
4. **State Transition Logic:** Combinational always block determines next state based on current state and counter.
5. **Output Logic:** Assigns traffic light, pedestrian, and buzzer outputs based on FSM state.
6. **Monitoring:** Testbench uses `$monitor` to display all states, counters, and outputs for simulation and debugging.

Conclusion

This FSM-based T-junction traffic controller meets all assignment requirements: independent traffic flows, pedestrian safety, and emergency vehicle bypass. The design is modular, safe, and easily extendable. Simulation and waveform analysis confirm correct operation under all specified scenarios.